

Ye (Lennon) Li, PhD, P.Stat.

Biostatistician | AI Workflow Architect | Public Health Data Scientist

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Profile

PhD biostatistician and Professional Statistician with 15+ years of applied public-health experience, 45+ publications/scholarly outputs, and 20+ graduate trainees supervised. **Current research focuses on trustworthy agentic data science: privacy-aware AI access to health data, typed decision models, probability calibration and uncertainty, human-gated automation, reproducible R/data systems, and statistical evaluation of AI outputs.** Combines statistical methodology, computer science, public-health domain knowledge, software engineering, and AI workflow architecture.

Core expertise

AI/agentic data science; statistical calibration and uncertainty; privacy-aware data access; human-in-the-loop systems; biostatistics; spatial/spatiotemporal statistics; surveillance and aberration detection; forecasting; R/Shiny/package development; reproducible research; graduate education.

Current appointments

Biostatistical Specialist — Public Health Ontario | Jan 2012–present

- Lead complex statistical analyses with scientists and epidemiologists; develop surveillance models, pipelines, interactive tools, and decision-support workflows for public-health evidence generation.
- Develop temporal, spatial and spatiotemporal methods for surveillance and aberration/cluster detection; advise on study design, modelling, uncertainty, interpretation, reproducibility, and statistical communication.

Adjunct Professor — University of Toronto, Dalla Lana School of Public Health, Division of Biostatistics | 2012–present

- Graduate teaching, practicum/thesis supervision, and mentorship in biostatistics and applied public health, emphasizing statistical reasoning, assumptions, uncertainty, and responsible interpretation.

Statistical Consultant & AI Workflow Architect — Biostats.ai | Jun 2010–present

- Design secure, reproducible, human-gated workflows in which AI agents assist with planning, coding, retrieval, execution, reporting, and documentation while people retain responsibility for assumptions, interpretation, approval, and release.

AI, data safety and open-source contributions

- **DataGangeR (CRAN/GitHub)** — Human-gated privacy protocol for agent-assisted prototyping. Produces synthetic stand-ins and reproducible generation recipes so coding agents can work without direct access to original records; designed around data minimization and no-network defaults.
- **Which (in development)** — Engine-neutral framework for typed AI decisions, probability normalization, calibration, abstention, uncertainty, cross-model comparison, regression release gates, and human-governed labelled-data/model-improvement workflows; current work includes Laya/Jev evaluation.
- **aiR** — Deployed AI-assisted companion for R statistical analysis with agent-guided exploration, modelling and interpretation while preserving analyst control over decisions and execution.
- **Rock** — Containerized R/data-science environment for Claude Code, Codex and Gemini CLI; isolates agents from host files and credentials by default and preserves human review before release.
- **insideR / IPD** — Transparency and deliberation tools for tracing/replaying R calls and for independent multi-agent review with preserved disagreement, uncertainty, and explicit human decision authority.
- **ONgeoR / OPCC / courieR** — Open-source R packages for Ontario public-health geography, source-qualified postal-code correspondence, and reproducible R package migration; developed with AI coding agents under human direction and review.

Current research

- **Privacy-aware AI data access** — calibrated statistical and semantic decisions about the minimum appropriate representation of medical/public-health data to expose to AI, with abstention, dual-gate safety, harm-weighted evaluation, and human review.
- **Typed-decision model evaluation** — converting free-form AI judgment into standardized classifications and probabilities that can be scored, calibrated, stress-tested, compared across models, and monitored over time.
- **PAGe infectious-disease forecasting** — phase-aligned influenza forecasting with ignition detection, epidemic-curve alignment/peak tracking, and short-horizon probabilistic forecasting.
- **Human-gated scientific agents** — multi-agent research/software workflows emphasizing delegation, independent verification, reproducibility, explicit uncertainty, controlled data boundaries, and accountable release decisions.

Education and credentials

- **PhD, Biostatistics, University of Toronto, 2012** — Division of Biostatistics, Dalla Lana School of Public Health. Dissertation: *Disease Mapping with Log Gaussian Cox Processes*.
- **MSc, Biostatistics, University of Toronto, 2007** — Department of Public Health Sciences, Faculty of Medicine.
- **Honours BSc with Distinction, Applied Statistics and Computer Science, University of Toronto, 2005**.
- **Professional Statistician (P.Stat.)**, Statistical Society of Canada — accredited since 2012.

Teaching and supervision

- 20+ graduate trainees supervised; longstanding practicum supervision for University of Toronto MSc/PhD Biostatistics students.
- Courses include CHL5201 Biostatistics for Epidemiologists I; CHL7001 Applied Spatial Statistics for Public Health Data; CHL5207/5208 Laboratory in Statistical Design and Analysis.
- Current educational focus includes statistical evaluation of AI classifications, probabilities, calibration, uncertainty, failure modes, and human-review thresholds.

Selected publications

- Ge E, Yu C, Lavigne E, Villeneuve PJ, Liu X, Grubic N, Lou W, Brook J, Lu Z, Li YL, To T. Early-life greenness and childhood asthma and allergic rhinitis: An Ontario birth cohort study. *European Respiratory Journal*. 2026. doi:10.1183/13993003.02272-2025.
- Peci A, Zhang P, Sullivan A, Li Y, Bi Y, Murphy A, Leibson K, Gubbay JB, Majury A. Impact of COVID-19 pandemic on Legionella testing and infection rates in Ontario. *BMC Public Health*. 2025;25(1):1-10.
- Paul LA, Li Y, Leece P, Gomes T, Bayoumi MA, Herring J, Murray R, Brown P. Identifying the changing age distribution of opioid-related mortality with high-frequency data. *PLoS ONE*. 2022;17(4):e0265509.
- Wilson SE, Bunko A, Johnson S, et al. The geographic distribution of un-immunized children in Ontario, Canada: hotspot detection using Bayesian spatial analysis. *Vaccine*. 2021;39(8):1349-1357.
- Li Y, Whelan M, Hobbs L, Fan WQ, Fung C, Wong K, Marchand-Austin A, Badiani T, Johnson I. Data Driven Approach of CUSUM Algorithm in Temporal Aberrant Event Detection Using Interactive Web Applications. *Canadian Journal of Public Health*. 2016;107(1):e9-e15.
- Li Y, Brown P, Rue H, Gesink D. Log Gaussian Cox processes and spatially aggregated disease incidence data. *Statistical Methods in Medical Research*. 2012;21(5):479-507.
- Li Y, Brown P, Rue H. Spatial modelling of lupus incidence over 40 years with changes in census areas. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*. 2012;61(1):99-115.

Technical profile

Statistics: R, Bayesian/spatial modelling, GAMs/regression, surveillance, forecasting, calibration, uncertainty, simulation, reproducible validation.

AI/systems: LLM/agent orchestration, typed decisions, model evaluation, human-in-the-loop control, privacy/data boundaries, local-vs-cloud execution.

Software: R packages, Shiny, Quarto, Python, APIs, containers, Git/GitHub, CI/CD, cloud deployment.

Links

Biostats.ai: <https://www.biostats.ai>

University of Toronto faculty profile: <https://www.dlsph.utoronto.ca/faculty-profile/li-lennon/>

GitHub: <https://github.com/lennon-li>